

OTOPAIR

INTERNAL TOOL SPECIFICATION · DOCUMENT 3 OF 3 · SECOND EDITION

Data Portal v2

The complete operating console for Otopair's vehicle-data asset: enrichment pipeline control, labor-time and parts-pricing command centers, tire intelligence, the Vehicle ID passport system, provenance governance, and the measured foundation for one day operating as a vehicle database

company. Every surface in this document traces to a dated, locked team decision.

VERSION	v2.0 — Build Specification (supersedes v1 + Appendix A)	DATE	July 11, 2026
BUILD OWNERS	Daniel C. (UI) · Waleed M. (pipeline queries) · Temur (director actions)	DOMAIN OWNERS	Waleed (pipeline) · Yassin (pricing, QA, catalog)
GROUNDING	V8 pipeline handoff · meeting notes Mar 23–Jun 22 · live production + ahmad (measured Jul 11)	STATUS	Ready for sprint scoping

1 Purpose, Thesis & What Changed in v2

Otopair is building two companies at once: a service marketplace, and underneath it a **proprietary vehicle-data asset that gets more valuable with every booking**. This portal is where the second company becomes visible, navigable, and governable — and where, one day, it becomes a product. v1 specified the browse-and-govern layer. v2 goes the rest of the way: it absorbs **every data-relevant decision the team has locked since March** and turns each into an operable surface. The difference is the difference between a viewer and a control room.

1.1 What v2 adds (each mapped to its source decision)

NEW IN V2	GROUNDING IN
Pipeline Control Room — batch-stage visibility (1A/1B/2), vin_queue management, per-VIN re-enrich & fix-prices triggers	V8 handoff (Mar 23) · "manual backfill / price-fix button exposed in the director page, per VIN" (Jun 4) · "director panel keeps a manual re-fetch / report-wrong-data button" (Jun 7) · live vin_queue = 584
SLO tiles with hard alert thresholds + push	Pipeline monitoring spec: success >80% (alert <70%), confidence >0.75 (<0.65), review queue <50 (>100), variance <5% (>10%), job confirmation >90% (<80%)
Labor Times Command Center	VD labor too low — CRV oil 18–25 min vs ~45 real; BMW battery 30 min vs 1h15+ (May 29) · RepairPal/MOTOR = high-confidence source; multipliers are guardrails, never truth (Jun 18) · RepairPal labor/quotes table shipped (Jun 22) · labor uniform car-to-car, not state-to-state (May 29)
Parts Pricing Engine console — median + outlier rejection, range caps, extraction-pathology flags	Pricing working session (May 28): median+outlier engine confirmed live (47/50/51.8/55.8 + \$90 outlier → \$51.19); 25% spread killed → 5–8% cap; single price → ±8%; price_cost_low/high stored but removed from math; "you save \$" / 4-pack / wrong-qty scraper pathologies
Pricing Engine (tiers & multipliers) INTERNAL	Tier multiplier built off one perfectly-populated Camry, T1 → T4, fallback + sanity-check only, never customer-facing (May 28) · build strictly to spec, director-side adjustment lever (Jun 7) · +50% brake/rotor quote cap (Jun 18) · live pricing_* family (11 tables, ahmad)
Tire Intelligence section	Tiered selection Elite/Mid/Standard; brand lists on our servers; rim diameter sufficient; wheelsize.com; tire data under trims (Apr 18) · 272 brands + parent-company map; 3 scrapers (SimpleTire, Walmart, Tire Rack); 94% fill on BMW 320i (Apr checkins) · price ceiling = scraped market data (Apr 20/23) · aftermarket = custom button, shop-confirmed, no quote (Jun 22)
Vehicle ID — passports & shop truth	Two-layer Vehicle ID: Layer 1 static AI-enriched read-only; Layer 2 living shop-modifiable (Apr 18 p.m.) · data hierarchy shop > AI > user, refined (Apr 18) · pre/post-job surveys, mandatory spec verification, "unchanged" toggle (Apr 18) · mechanic-review inspection fields: tread depth in 32nds, rotor thickness micrometer + descriptors, green/yellow/red gradient (Jun 19) · live vehicle_passports = 33 (ahmad)
Consensus & evidence mechanics, verbatim	V8 handoff: score = 0.4·avgConf + 0.3·(sources/total) + 0.3·maxConf; mechanic boost ×1.2 capped 1.0; conflict if runner-up >0.5; normalization per field family; is_latest semantics; verification confidences 0.98/0.99; empirical promotion at 3+ samples; config "verified" at 3+ verifications
Source & scrape governance, verbatim rules	Auto-block <40% accuracy after 20+ observations (V8) / "<40% across 3 vehicles → blocked via backend CRM-style panel, manual override" (Apr 2) · reliability = accuracy·0.7 + min(obs,100)/100·0.3 · cache TTLs 30/90/7 days · URL-pattern registry (partsdeal, oempartsonline) with fallback chain
Service catalog cleanup as a tracked task	"App shows only the 23 established services across 4 categories; clean the database so search can't surface phantom services" (Jun 22) — which resolves the 7-vs-4 category drift measured live in production on Jul 11
VD dump & credit economics	"Use the 8GB DB dump locally instead of per-call API" + parts pricing NOT sourced from VD, labor only (May 29) · credit-tier discussion (Apr 13) · \$7,000 / 36K credits contract

Data-company foundation (Appendix B)	The internal → external console thesis; dual-source pipeline "VD + AI enrichment as equal peers" (Apr checkins); flywheel gates
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1.2 Design laws (unchanged, hard-won)

- **Provenance is a first-class citizen.** Every field everywhere renders its layer badge + confidence. Ford-on-Alfa happened because layers trusted each other implicitly; this portal makes trust explicit and visual, always.
- **Shop truth outranks everything.** The locked hierarchy — shop > AI > user for enrichment-derived fields, shop > user > AI for user-provided fields (Apr 18) — is rendered, enforced, and never silently violated.
- **Every read view is built as if a paying customer will see it.** Same components, different auth, later.
- **Reads are free, writes are ceremonies.** Reason required, before → after diff, one `audit_log` row atomically (Apr 20 decision; table confirmed live Jul 11).

2 Users & Permissions

PERSONA	WHO	FOCUS
Super Admin	Temur, AB	Everything; pricing-lever changes; incident declaration; catalog-wide mutations.
Data Admin	Waleed, Yassin	Pipeline triggers, source blocking, review resolution, labor/pricing configuration, tier tables.
Reviewer	Yassin + trained members	Review queue throughput; approve/reject/edit with evidence. No pipeline or pricing powers.
Read-only	Ahmad, Daniel, Bilal	Full visibility. Ahmad debugs what the app renders per VIN; Daniel traces shop-reported specs.

CAPABILITY	SUPER	DATA ADMIN	REVIEWER	READ
Browse everything (catalog, pipeline, pricing, tires, passports, costs)	✓	✓	✓	✓
Approve / reject / edit review items	✓	✓	✓	—
Trigger re-enrich / fix-prices per VIN; manage vin_queue; clear cache	✓	✓	—	—
Block/unblock sources (manual override of auto-block)	✓	✓	—	—
Edit labor-source weights, range caps, quote caps (the "director adjustment lever," Jun 7)	✓	✓ (co-sign)	—	—
Edit tier tables / multipliers / Camry base INTERNAL	✓	Yassin ✓ (co-sign)	—	—
Edit service catalog; run 7 → 4 category cleanup	✓	✓	—	—
Declare / resolve data incident	✓	✓	—	—

Shared shell

Identical to Doc 1 §4: ⌘K, entity chips, write ceremony → `audit_log`, four states per page, index-only pagination, environment banner (critical: ahmad is seed-rich and is *not* production — measured, not folklore).

3 Provenance Taxonomy — now with the exact source_type mapping

LAYER	ORIGIN	MAPS FROM ENRICHMENT_EVIDENCE.SOURCE_TYPE	LICENSE POSTURE
A	NHTSA vPIC decode, config identity	nhtsa	Public domain — clean
B	Vehicle Databases (API + the locally-hosted 8GB dump)	vehicle_databases (source tag to be confirmed present – see §12 risk)	Blocked externally pending contract review. Labor-time consumption only; parts pricing explicitly NOT sourced from VD (May 29)
C	Scraped retail/spec/manual sources	scraped, web_search, gap_fill	Internal use; non-sellable v1
D	Mechanic-verified values, corrections, empirical labor, shop_vehicle_data, pre/post-job survey truth, labor tiers, catalog structure, consensus metadata	mechanic	Clean — this is the product
E	Model-extracted / inferred without independent source	training_data, sibling_engine	Usable; weak to sell alone
X	Untraceable / ambiguous	—	Excluded externally; queued for backfill

Layer is computed on the read path — never hand-assigned; ambiguity renders X. RepairPal/MOTOR-derived labor values are their own sub-tag within C/B pending license reading (MOTOR sandbox doc on file; not contracted). The badge system must distinguish repairpal_motor explicitly because it is now the high-confidence labor source (Jun 18) with its own legal posture.

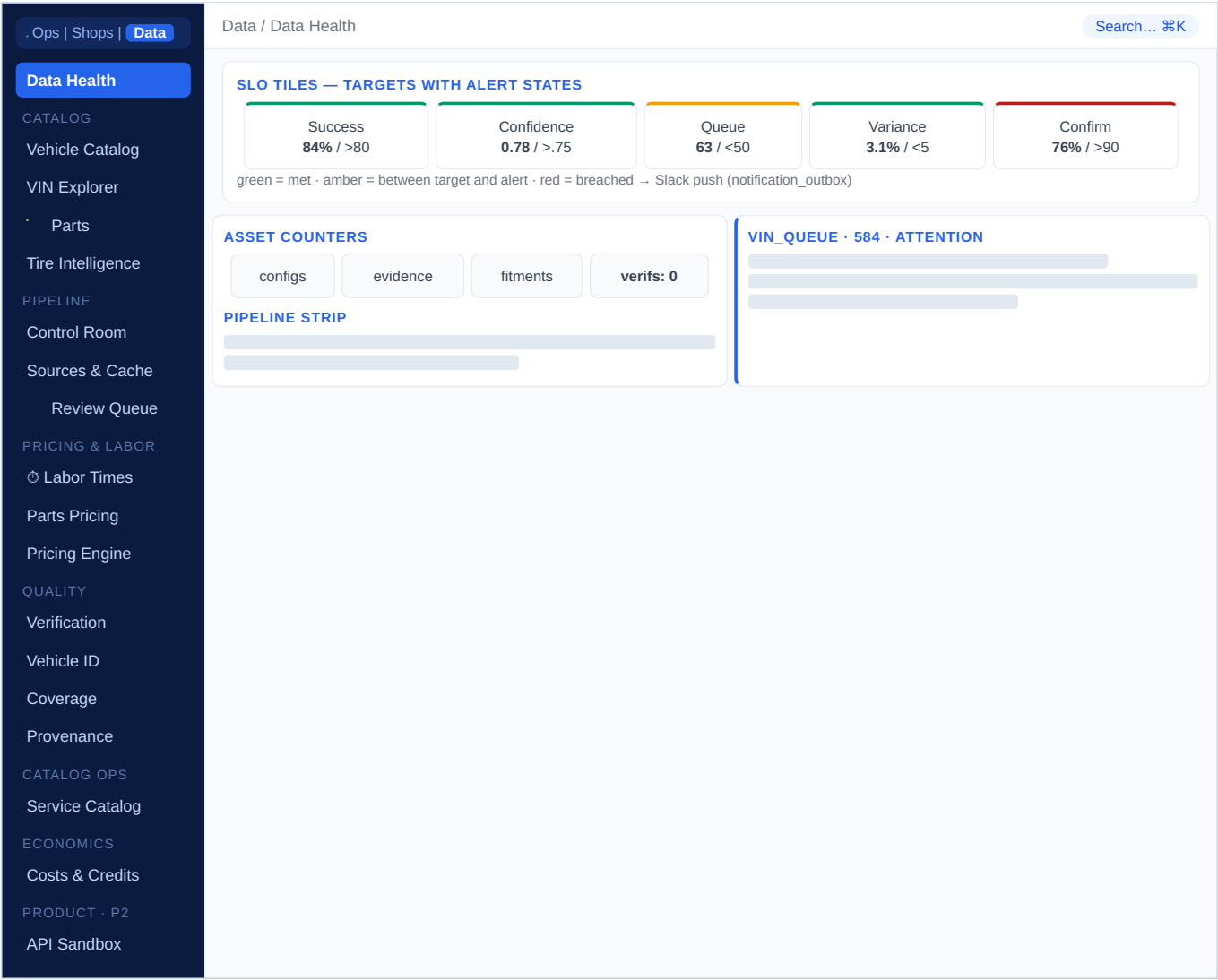
4 Information Architecture v2

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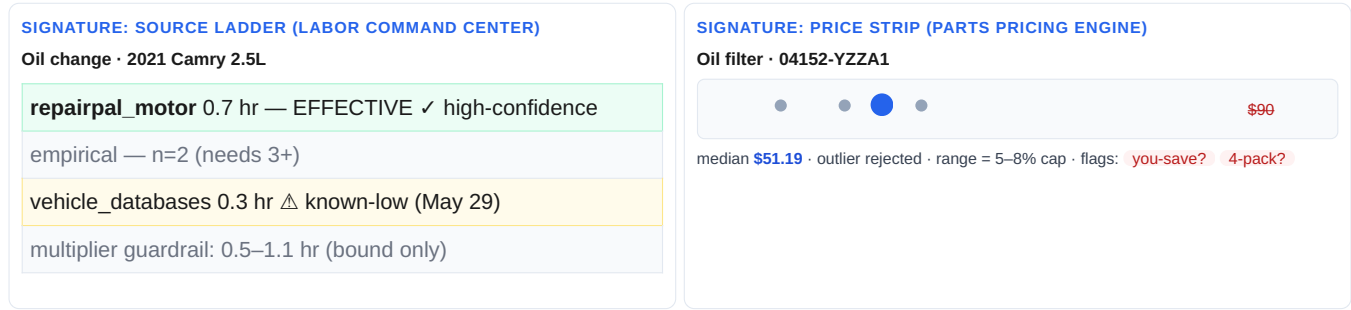
DATA – portal switcher: [ Ops | Shops | Data ]
├─ Data Health Overview /data (SLO tiles + alerts)
├─ CATALOG
│   ├── Vehicle Catalog /data/catalog (make-model-trim-config workspace)
│   ├── VIN Explorer /data/vins · /data/vins/:vin
│   ├── Parts /data/parts · /data/parts/:oem
│   └─ Tire Intelligence /data/tires (brands · models · pricing · sizes)
├─ PIPELINE
│   ├── Control Room /data/pipeline (runs · batches · vin_queue · triggers)
│   ├── Sources & Cache /data/sources
│   └─ Review Queue /data/review
├─ PRICING & LABOR
│   ├── Labor Times Command Center /data/labor
│   ├── Parts Pricing Engine /data/parts-pricing
│   └─ Pricing Engine (tiers) /data/pricing [INTERNAL – gated]
├─ QUALITY
│   ├── Verification & Variance /data/verification
│   ├── Vehicle ID – Passports /data/passports (shop truth · surveys)
│   ├── Coverage & Quality /data/coverage
│   └─ Provenance & Incidents /data/provenance
├─ CATALOG OPS
│   └─ Service Catalog /data/services (incl. 7-4 cleanup task)
├─ ECONOMICS
│   └─ Costs & Credits /data/costs
└─ PRODUCT (Phase 2)
    └─ API Sandbox /data/api
  
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4A Screen Anatomy — Data sidebar & signature layouts

Shell and templates T1–T5 per Doc 1 §3A. The Data portal adds two signature layouts of its own: the **source ladder** (Labor Command Center) and the **price strip** (Parts Pricing Engine), both wireframed below with the exact sidebar.



The Data shell at /data — five SLO tiles with live target math, moat counter rendered honestly.



The two layouts the May 28 / Jun 18 decisions earned — rendered, not described.

4B Page Layout Atlas — every Data page, top to bottom

Data Health Overview T1+SLO · sidebar item 1 · /data	
SLO row	5 tiles per wireframe: value 26px, target line 12px "84% / target >80%", 3px top border = state color; red tile shows " → pushed to #otopair-db 09:14".
Second row	Left: asset counters (verifications rendered largest, 32px, at its honest value). Right: pipeline strip — runs 7d, cost/run vs \$0.30–0.60 band (mini bar with band shading), FireCrawl burn, VD meter n/36,000.
Third row	Left: vin_queue card — 584, age histogram 5 bars, drain projection line 13px. Right: attention list (failed runs / near-block sources / X-layer delta / stale reviews / open incidents).

Vehicle Catalog drill + T3 workspace · CATALOG → Vehicle Catalog · /data/catalog	
Drill levels	Makes = logo grid cards (fill-ring 36px each) → Models table → Trims table (year span, variant counts, fill %) → Config workspace.
Workspace header	"2021 Toyota Camry · 2.5L A25A-FKS" 20px · fill ring · verification count chip · "N real VINs on platform" chip · actions right: Re-enrich · Flag · Export JSON.
Body grid	2-col: left — Variants selector (engines/transmissions/chassis segmented rows), then Specs card: every field = label 12px #6B7280 · value 14px · layer badge (A blue / B grey / C amber / D green / E purple / X red, 10px pill) · confidence 11px · history icon. Right — Fitments card grouped by service_type (live shape: config+part+service), conflict rows red-bordered; Intervals card; Service math card (est vs empirical per service, red INTERNAL tag).
Field history	Drawer: evidence rows grouped by normalized value, group score bars, winner ★, mechanic ×1.2 badge where present.

VIN Explorer T2 → T3 · CATALOG → VIN Explorer · /data/vins	
List	VIN mono 13px · year · make/model/trim chips · engine code · image ✓ · owners · bookings · "missing links" filter default off but pinned (trim_id/engine_id null = decode gaps, amber rows).
Detail	Left: decode card (all FKs as chips) + metadata JSON viewer (collapsed) + Re-decode ghost button. Right: vehicle image card with "IMAGIN — licensed, never export" 11px footer; passport link card; ownership (count + dates only, UserChip gated); booking chips.

Parts T2 → T3 · CATALOG → Parts · /data/parts	
List	OEM# mono · name · category pill · fitment count · hygiene filters: Unnamed · Unfitted (orphans).
Detail	Header: part identity + category. Center: reverse fitment graph — grouped-by-make expandable rows, each config chip → workspace; count banner "correction touches N configs" (the blast-radius view). Bottom: usage panel — job_actuals appearances with cost distribution mini-histogram (Layer D price signal).

Tire Intelligence T3 · 4 tabs · CATALOG → Tires · /data/tires	
Tabs	Brands · Models & Pricing · Sizes · Quote Guardrails.
Brands	Table: brand · tier pill (Elite gold-bordered / Mid blue / Standard grey) · parent company · scrubbed ✓; pinned note card: "brand-level tiering is the MVP call (Goodyear Eagle F1 caveat) — refine post-launch."
Models & Pricing	tire_models table with price fill % column vs 94% benchmark; prices badged "ceiling reference"; source chips (SimpleTire / Walmart / Tire Rack) with per-source freshness dots (7d TTL).
Sizes	Per-trim OEM sizes, staggered F/R rendered as two chips, rim diameter only; wheelsize.com status card (calls today / 300 free).
Guardrails	Ceiling-breach quote list (quote chip → shop) + aftermarket flags ("custom size — shop-confirmed, no quote").

Pipeline Control Room T3 · 3 tabs · PIPELINE → Control Room · /data/pipeline

Runs tab	Run rows: trigger pill · stage chip rail (cache → scrape → 1A 1B → merge → checks → B2 → write, done=green / active=pulsing blue / failed=red with error count) · duration · tokens · credits · cost vs band · fill delta bar · cache_hit badge. Row expand: per-stage timings table + batch IDs mono with copy + error texts.
VIN Queue tab	Header stat trio: backlog 584 · throughput/day · projected drain date. Table: VIN · make/model when known · age band pill · attempts · last error (clamp) · priority; bulk bar appears on multi-select (Retry · Bump · Remove); Dead-letter sub-view (≥3 fails) grey-tinted.
Triggers tab	Three action cards with VIN input each: Re-enrich · Fix prices · Report wrong data — each shows cooldown state ("available in 42m"), last-run link, cost estimate; Bulk trigger card (Super Admin) with count+cost preview before confirm.

Sources & Cache T3 · 4 tabs · PIPELINE → Sources · /data/sources

Registry	Domain · observations · accuracy sparkline with 40% red threshold line · reliability score (formula on hover) · status; "nearing block" filter (40–50%, 15+ obs) default-pinned.
Blocked	6 rows: domain · reason · blocked_by pill (manual/auto_accuracy/seed) · accuracy_at_block · Unblock (ceremony).
URL Patterns	Per-make pattern cards: template mono 12px · slug list chips · fallback chain diagram (year → generic → skip) · measured success %.
Cache	By source_type with TTL badges (30/90/7d) · expiring-soon list · hit-rate 30d stat · per-vehicle Clear (ceremony, cost warning).

Review Queue T4 board → T5 workspace · PIPELINE → Review Queue · /data/review

Board	Pending → In Review → Resolved; card = priority pill · stream badge (consensus/correction/report/survey) · entity chips · age · assignee avatar.
Workspace	T5 three-pane per the wireframe: queue rail (↑↓) · center evidence panel — value groups as scored bars (0.4/0.3/0.3 math on hover, mechanic ×1.2 badge, winner ★, runner-up score visible) with source URLs opening beside · right context (live value + layer, affected VINs, variances, workspace link). Footer keys: Approve Reject Edit.
Throughput strip	Resolved 7d · median TTR · approval % · per-reviewer chips — vs the <50 depth SLO.

Labor Times Command Center T3 · 5 panels · PRICING & LABOR → Labor Times · /data/labor

Selector bar	Service ▼ × Config search — everything below scopes to the pair.
Source ladder	The signature layout (wireframed above): one row per source, effective value green-highlighted with selection reason; VD rows permanently carry the amber "known-low (May 29)" badge.
Test bench	T1 → T4 reference cars as columns × sources as rows; deviation-from-RepairPal heat coloring; pinned regression cards (CRV oil · BMW battery) top-right, green until any source reproduces them.
Breaches	Guardrail-bound violations + brake/rotor +50% cap trips, each with quote/job chip.
Empirical ledger	labor_observations table: service×config · n · mean · p25/p75 (greyed n<5 "insufficient sample") · growth sparkline.
Config panel	The director lever: source weights · caps · thresholds — Super Admin+co-sign, history list beneath, "uniform car-to-car, not state-to-state" fixed banner.

Parts Pricing Engine T2 → strip detail · PRICING & LABOR → Parts Pricing · /data/parts-pricing	
Part strip	The signature price strip (wireframed above): source dots on a horizontal scale, outlier struck red, median 14px blue, computed range bracket per rule (5–8% multi / ±8% single / multiplier badge zero-source).
Pathology flags	Red chips per suspect capture (you-save / multi-pack / qty); click = split view source page ↔ captured value with Confirm pathology / Legit buttons; confirms feed source scoring.
Legacy note	price_cost_low/high columns greyed with tooltip "stored for records — removed from math (May 28)."
Validation queue	RepairPal parts tier-test runs listed with results; source-weight change proposals attach the test as evidence (ceremony).

Pricing Engine T3 gated · PRICING & LABOR → Pricing Engine · /data/pricing · INTERNAL	
Gate	Interstitial: red-bordered card "Internal pricing system — access logged" + re-auth; page views write audit rows.
Layout	Left: tier list T1 → T4 with vehicle-assignment counts (300-model list searchable). Center: Camry golden record — completeness ring 64px + per-service base values. Right: multiplier matrices (labor/parts × category) as editable grids, edit = ceremony + co-sign + blast-radius preview ("affects N configs' fallback").
Banner	Fixed verbatim policy strip: "Fallback & sanity-check only — never customer-facing, never overrides a sourced price."

Verification & Variance T3 · 4 tabs · QUALITY → Verification · /data/verification	
Verifications	Event stream: mechanic chip · config · per-field verdict chips (confirmed green 0.98 / corrected amber 0.99 showing old → new) · milestone chips (" → empirical at 3 samples", "config verified") rendered as small gold badges.
Variances	Flagged-first table: predicted vs actual columns with delta bar between · job chip · reviewed toggle; header medians by service / by make.
Confirmations / Empirical	Confirmation rows with feedback text clamp; Empirical = shared ledger view from Labor CC.

Vehicle ID — Passports T2 → two-layer detail · QUALITY → Vehicle ID · /data/passports	
List	VIN chip · vehicle · two completeness rings (L1 / L2) · last shop touch · verified fields count.
Layer 1 region	Slate-tinted card #F8FAFC with lock icon header "Static — AI-enriched, shops cannot edit": VIN, make, model, year, trim, engine, drivetrain as read-only rows; zero edit affordances by construction.
Layer 2 region	White card "Living — shop truth": each field row = current value 14px · setter shop chip · date · hierarchy ladder mini-diagram (shop > AI > user or shop > user > AI per field class) · timeline expander; override events render as timeline milestones "shop value ×3 consistent → overrode research (date)".
Surveys	Pre-job / Post-job streams: submission cards linked to booking+mechanic; inspection fields native — tread 32nds per-corner 2×2 mini-grid, rotor thickness 3-decimal inputs + descriptor chips (scored/pitted/rusted/warped), G/Y/R gradient dots for fluids & filters, photo thumbnails 48px; per-shop red-dot compliance % chart at top.

Coverage · Provenance & Incidents matrix + T2 · QUALITY group	
Coverage	Make × year-band matrix, fill shading, applicability-aware denominators; below: per-field-family bars with the 35%/52% → 89% fallback reality annotated; launch-make tracker chips (green/amber).
Provenance	Two moat lines chart (D↑, X↓) · Layer B exposure table (VD-only fields, exportable for legal day).
Incidents	Cards: #1 Ford-on-Alfa (sev, 38 configs, root cause, per-deployment scope note) · #2 VD labor under-read; "Declare incident" from any page's flag menu pre-fills context.

Service Catalog

T3 + migration workspace

CATALOG OPS → Service Catalog · /data/services

Catalog	23 services grouped by category; row = name · slug mono · default_labor_hours (co-sign edit) · options count · drag handle; options editor drawer with live phone-frame preview.
Migration card	Pinned amber workspace "7 → 4 category cleanup (Jun 22, owner: Temur)": mapping table (7 live → 4 locked), phantom-service detector list (0 offerings + 0 bookings), Dry-run diff button → full change preview, Execute (ceremony); completion collapses the card to a green record with evidence link.

Costs & Credits · API Sandbox

T1 charts / T3

ECONOMICS / PRODUCT groups

Costs	Cost-per-run trend with \$0.30–0.60 band shading · token area chart · FireCrawl burn vs free-tier line · VD credit meter : radial n/36,000 + weekly burn + projected exhaustion date, annotated cards: "8GB dump hosting (owner Waleed) — replaces per-call charges" and "credit-tier downgrade question (Apr 13)".
Sandbox	Keys table (prefix-visible, scopes chips, revoke) · usage chart per key · Playground: config picker → JSON viewer with excluded fields listed under "blocked by layer B/C" in red strip · labor endpoint preview (empirical-only) · generated docs pane.

5 Data Health Overview — /data — SLOs with teeth

v1's overview showed numbers; v2's overview shows numbers **against locked targets, with alert states and push**. The five SLO tiles come straight from the pipeline monitoring spec:

SLO TILE	TARGET	ALERT IF	COMPUTED FROM
Enrichment success rate (7d)	> 80%	< 70%	enrichment_runs status complete/failed
Average confidence	> 0.75	< 0.65	consensus confidence across latest evidence
Review queue depth	< 50	> 100	queue table (\$8) pending count
Spec variance rate	< 5%	> 10%	spec_variances flagged ÷ jobs
Job confirmation rate	> 90%	< 80%	post-job survey confirmations ÷ completed jobs

Tile states: green (target met) · amber (between target and alert) · red (alert breached → one row written to notification_outbox → Slack #otopair-db, deduped per day). Below the tiles: **asset counters** (configs, evidence rows, fitments, verifications — the moat counter rendered biggest, honestly at its live value), **pipeline strip** (runs 7d, cost/run vs the \$0.30–0.60 target band, FireCrawl burn, VD credit meter /36,000), **vin_queue backlog** (584 live) with age distribution, and the **attention list** (failed runs 24h, sources nearing auto-block, X-layer growth, stale review items >72h, open incidents).

6 Pipeline Control Room — </data/pipeline>

6.1 Runs — every stage of every tier, visible

Tier 1 runs render their true internal shape from the V8 handoff, not a generic status: **cache check** → **identity** → **FireCrawl scrape (parts catalog + owner's manual)** → **Batch 1A (extract, no search)** || **Batch 1B (web search, ≤15 uses)** → **poll (60s × ≤180)** → **merge (1A wins)** → **applicability rules** → **sanity checks** → **OEM validation** → **Batch 2 (gap fill + 25-service pricing, ≤20 searches)** → **poll** → **write normalized tables** → **fill rate** → **complete**. Each run row: trigger (vin_decode / manual / test) · stage chips with per-stage timing · batch IDs (copyable) · tokens in/out · FireCrawl credits · cost vs target band · fill delta (fields_filled/total; fields_changed expands) · cache_hit badge · errors (expand, full text). Filters: status, trigger, make, date. **Tier 2 runs** show per-source site-scoped query counts, evidence rows added (batched by 50), and consensus CONFIRMED/CHANGED log lines. **Tier 3** events stream from mechanic verifications (\$10).

6.2 VIN Queue — the backlog is a workspace, not a number

vin_queue (584 live in production): VIN (mono) · resolved make/model chips when decodable · queued at · age band · attempts · last error · priority. Actions (Data Admin): bump priority, retry now, remove (ceremony), bulk retry by error class. Header: throughput/day, projected drain date at current rate, dead-letter count (≥3 failed attempts, quarantined view). This page answers the question the team currently answers by feeling: *"how far behind is enrichment, and is it draining or growing?"*

6.3 Triggers — the director levers, formalized

Three per-VIN actions, all locked decisions made real: **Re-enrich** (full Tier 1; queues run, links it back — the Jun 7 "manual re-fetch" button) · **Fix prices** (the Jun 4 backfill-per-VIN action; runs the parts-price backfill for one VIN and links the diff) · **Report wrong data** (creates a review item pre-loaded with the VIN's context — the Jun 7 report button; mechanic-side reports route here too, into the team review dashboard as decided). Every trigger: ceremony, audit row, and rate-limit (per-VIN cooldown 1h) so the button can't be mashed into a cost incident. Bulk trigger (by make/model/year) is Super Admin only, previews run-count and estimated cost before confirm.

7 Sources & Cache — </data/sources>

Registry ([source_registry](#), 158 live): domain · total_observations · accuracy_rate · reliability_score (accuracy×0.7 + min(obs,100)/100×0.3 — the formula rendered on hover, not hidden) · sparkline with the **40% red threshold line** · status.

Auto-block rule shown as law: accuracy < 40% after 20+ observations → blocked, row written to [blocked_domains](#) with blocked_by: auto_accuracy and accuracy_at_block (V8; the Apr 2 "CRM-style panel with manual override" is this page). Manual block/unblock = ceremony. **Blocked domains** tab: 6 live rows, reason + blocked_by (manual / auto_accuracy / seed).

URL patterns tab: the deterministic scrape registry — partsdeal.com family (BMW 14 slugs; Toyota/Honda 8) and oempartsonline.com subdomains (25 makes, 8 slugs), with the fallback chain (year-specific → generic → skip to Batch 2) documented per make and per-pattern success rate measured from scrape results. **Cache** tab: [scrape_cache](#) (40 live) by source_type with TTL badges (parts 30d / manuals 90d / pricing 7d), expiring-soon view, hit-rate 30d, and clear-cache per vehicle (maps to debugClearVehicleCache; ceremony — cache clears cost real money downstream).

8 Review Queue — `/data/review` — the quality production line

Backing-table decision (was v1's open item, now scoped): live production has no `manual_review_queue`; the queue is fed by four real streams — consensus rows where `needs_review = true` (conflict + no mechanic verification, per the V8 rule), mechanic corrections (Tier 3), "report wrong data" triggers (§6.3), and post-job survey disconfirmations. Waleed materializes these into one thin `review_queue` table (proposed §13) so assignment and SLA can exist; the streams stay the source of truth.

Board: Pending → In Review → Resolved; priority × age sort; assignment (me / teammate / bulk-by-make). **Workspace** (keyboard-first — A approve / R reject / E edit / ↑↓): left, the proposal with source + confidence; center, the **evidence panel** — all `enrichment_evidence` rows for the field, grouped by normalized value exactly as the consensus engine groups them, each group scored with the real formula ($0.4 \cdot \text{avgConf} + 0.3 \cdot \text{share} + 0.3 \cdot \text{maxConf}$, mechanic ×1.2 badge when present), winner highlighted, runner-up score shown so the reviewer sees *why* it conflicted; right, context — current live value + layer, affected real VINs count, related variances, config workspace link. **Decisions:** Approve (write, resolve, audit) · Reject (reason; optional auto re-enrich) · Edit-then-approve (manual value → evidence row `source_type` mechanic-equivalent "manual" at reviewer identity). **Throughput strip:** resolved 7d · median time-to-resolution · approval rate · per-reviewer counts — measured against the <50 depth SLO.

9 Pricing & Labor — the business-critical trio

9.1 Labor Times Command Center — </data/labor>

Why this page exists: wrong labor time is the one data error that directly costs a shop money on a live job ("price shows too low, customer books, shop eats the loss — and the gap falls back on us," May 29). It nearly moved the launch date. This page makes labor truth inspectable service-by-service, car-by-car.

PANEL	CONTENTS
Source ladder	Per service × config: every labor value we hold, one row per source — repairpal_motor (high-confidence, Jun 18) · vehicle_databases (known-low; badge carries the May 29 warning) · empirical (rolling average, sample_size shown; promoted at 3+ samples) · multiplier_guardrail (bound only, never truth — Jun 18). The <i>effective value</i> the app uses is highlighted with its selection reason.
Tier test bench	The Jun 18 comparison methodology as a permanent tool: pick a service, render values across the T1 → T4 reference cars side-by-side per source; deviations from RepairPal by-the-book flagged. Seeded with the canonical bad examples (CRV oil 18–25 min vs ~45; BMW battery 30 min vs 75–90 incl. mandatory post-diagnostic) as pinned regression cases — if a source ever reproduces them, the row lights up.
Guardrail breaches	Values falling outside multiplier bounds, and quotes breaching the +50% brake/rotor cap (Jun 18; confirm-vs-block behavior read from config). Each breach links to the job/quote that tripped it.
Empirical ledger	labor_observations (91 live, ahmad) + labor_times empirical fields: per service×config actuals, sample sizes, p25/median/p75 once n≥5 (greyed "insufficient sample" below). This ledger is the future sellable empirical layer; its growth is charted here.
Config	The director adjustment lever (Jun 7): source priority weights, cap values, promote/demote thresholds — Super Admin + co-sign, ceremony, full history. "Uniform car-to-car, not state-to-state" (May 29) rendered as a fixed banner so nobody re-adds a state dimension casually.

9.2 Parts Pricing Engine — </data/parts-pricing>

The May 28 engine, made visible. Per part (from [part_prices](#), 472 live): all sourced prices as dots on a strip → outlier struck through → **median highlighted** (the worked example 47 / 50 / 51.8 / 55.8 with \$90 rejected → \$51.19 ships as the empty-state illustration). Range logic per the locked rules: multi-source → 5–8% cap between ends (25% spread is dead); single source → ±8%; zero sources → Yassin multiplier fallback, badged **INTERNAL FALLBACK**. [price_cost_low/high](#) render greyed with the note "stored for records — removed from pricing math (May 28)". **Extraction pathology detector:** flags for the three known scraper failure modes — "you save \$X" captured as price · multi-pack total read as per-unit · wrong per-cylinder quantity — each flag opens the source page beside the captured value for a ten-second human verdict, and confirmed pathologies feed source scoring. **Validation queue:** the Jun 18 next-step (same tier test on part prices via the RepairPal endpoint) runs here; if it holds, source weights shift and scraping weight drops — the weight change is a ceremony with the test results attached as evidence.

9.3 Pricing Engine — tiers & multipliers — </data/pricing> **INTERNAL — GATED**

Extra auth gate (Super Admin + Yassin's Data Admin only; every view of this page is itself audit-logged). Surfaces the live [pricing_*](#) family: tiers (T1 → T4) with vehicle assignments (the 300-model list), per-tier per-service multipliers, the **Camry base car** — one perfectly-populated reference whose values everything derives from (May 28) — rendered as a golden record with completeness ring; baselines, labor/parts category multipliers, fallback snapshots. Edit = ceremony + co-sign; every change previews blast radius ("affects N configs' fallback prices"). Fixed banner, verbatim policy: *fallback and sanity-check only — never customer-facing, never the source of truth over a sourced price*. Built strictly to spec (Jun 7); the adjustment lever lives here and nowhere else.

10 Quality Surfaces

10.1 Tire Intelligence — /data/tires

Brands: [tire_brands](#) (45 live production; 272 tiered in the full set) — brand · tier (Elite / Mid-range / Standard, assigned by brand quality not price point) · parent company (the sub-brand map) · Yassin-scrubbed badge (trucks/motorcycles/non-USA purged). Tier edits = ceremony; the known vulnerability is pinned as a standing note: *same brand spans quality levels* (Goodyear Eagle F1 = Elite quality, Mid brand) — *brand-level tiering is the MVP call, refine post-launch*. **Models & pricing:** [tire_models](#) (813) · [tire_pricing](#) (1,036) — scraped from the three-source pipeline (SimpleTire, Walmart, Tire Rack; parallel, cache-first), price fill % per size tracked against the 94% benchmark; prices render with their role stated: **ceiling reference for quotes, quality tier is the signal** (Apr 20/23). **Sizes:** per-trim OEM sizes (tire data lives under trims — Apr 18), staggered front/rear handling, rim diameter (the only rim spec in scope), [tire_size_cache](#) health, wheelsize.com source status (300 free calls/day; \$500/yr production decision pending usage). **Quote guardrails:** ceiling breaches by shop quotes listed with quote chips; aftermarket sizes flagged as custom-button/shop-confirmed/no-quote per Jun 22.

10.2 Vehicle ID — Passports & Shop Truth — /data/passports

The persistent car "license" (Apr 18), operable. Passport list ([vehicle_passports](#), 33 live ahmad): VIN chip · vehicle · completeness ring per layer · last shop touch · verified-fields count. **Passport detail, two layers rendered as two visually distinct regions:**

- **Layer 1 — static, read-only:** VIN, make, model, year, trim, engine, drivetrain from AI enrichment. Shops cannot edit (Apr 18 p.m. — protects against mechanic error); edit affordances simply don't exist here, with a lock explainer.
- **Layer 2 — living:** tires, fluids, brakes, mileage, mods, mechanic notes from [shop_vehicle_data](#) / passport rows. Every value shows: current, who set it (shop chip → Shops portal), when, prior values timeline, and consensus state under the locked rule — **shop-entered 2–3× consistently overrides research data**; the override moment is rendered as an event, not a silent swap.

Hierarchy visual: shop > AI > user for enrichment-derived fields; shop > user > AI for user-provided (mileage, inspection date) — drawn as a small ladder on every contested field. **Survey streams:** pre-job (mandatory onboarding: tire specs, oil, coolant, brake fluid, part numbers, mileage; "unchanged" toggles for repeat vehicles) and post-job (mandatory spec verification; optional optimization questions; N/A for untouched services) — each submission is a row linked to booking + mechanic, with the Jun 19 inspection instrumentation rendered natively: **tread depth in 32nds per corner, rotor thickness (3-decimal inches / microns — unit lock pending), rotor descriptors multi-select (scored / pitted / rusted / warped), green-yellow-red gradient for fluids & filters, photo attachments**. Service-specific required-field (red-dot) compliance rate per shop is charted — the "forced measurements" Abd endorsed become a measurable quality metric per partner.

10.3 Verification & Variance — /data/verification

Four tabs, v1 structure retained, mechanics deepened: **Verifications** — Tier 3 stream with the exact write semantics displayed per event: confirmed → evidence at 0.98; corrected → prior evidence `is_latest=false` + corrected at 0.99; actual hours → empirical rolling average, `sample_size++`; promotion notices ("source → empirical at 3+ samples", "config → verified at 3+ verifications") render as milestone chips — the flywheel's gear-teeth, visible. **Variances** — [spec_variances](#) flagged-first, predicted vs actual, median by service and by make. **Confirmations** — post-job spec confirmations; disconfirmations auto-queue. **Empirical** — the labor ledger view shared with §9.1. Live counter honesty: verifications = 0 today, rendered at 0.

10.4 Coverage & Quality — [/data/coverage](#) · Provenance & Incidents — [/data/provenance](#)

Coverage: make × year matrix with fill shading; per-field-family completeness with the launch reality pinned — intervals 35% / labor 52% pre-fallback, 89% where fallback applies (Apr measurement) — tracked as a trend so the Saturday data sessions (Apr 2 decision) aim at the true weak spots; applicability-aware denominators (EV/ICE, timing belt, staggered — the six V8 applicability checks) so a field that shouldn't exist never counts as missing. **Provenance:** layer distribution over time (D growing? X shrinking? — the two moat lines); Layer B exposure report (VD-only-evidence fields, pre-computed for legal-review day); **Incidents:** [data_incidents](#) (proposed) seeded with Incident #1 Ford-on-Alfa (38 configs, clone-injection root cause, deployment-scoped — production holds only 30 configs total, so backfill scope is per-deployment by measurement) and Incident #2 VD labor-time under-read (May 29, root cause pending Saad's dev). Institutional memory lives in the tool, not chat scrollback.

11 Service Catalog — [/data/services](#) · Costs & Credits — [/data/costs](#)

Service Catalog (Yassin's domain): 23 services (count verified live ✓) with slug, category, default_labor_hours (ceremony + co-sign; multiplies across every shop's pricing), options editor (52 option rows live in ahmad) with phone-frame preview, category manager. **Front-and-center tracked task — the 7 → 4 cleanup:** production holds 7 categories (Diagnostics, Compliance, Routine Maintenance, Tires, Brakes, Battery, Fluids — measured Jul 11) versus the Jun 22 locked decision: *app shows only the 23 established services across 4 categories; clean the database so search can't surface phantom services*. The page ships with a one-time migration workspace: category mapping table (7 → 4), phantom-service detector (services with zero shop offerings and zero bookings), dry-run diff, execute-with-ceremony, owner Temur per the Jun 22 assignment. The drift measured in Appendix A wasn't drift — it was this decision not yet executed; the portal turns it from a Slack memory into a completable task with evidence.

Costs & Credits: cost-per-run trend vs the \$0.30–0.60 band (production-ready benchmark \$0.30/VIN); token in/out from runs; FireCrawl credit burn vs free tier; **VD credit meter** — used / 36,000 with weekly burn and projected exhaustion, annotated with the two standing economics decisions: the 8GB local dump replaces per-call charges once hosted (May 29 — hosting owner Waleed, status tracked here), and the credit-tier downgrade question (Apr 13, "we won't burn 120 in year one" — this graph is the evidence for the renewal call). Batch-API savings line (50% vs realtime, per V8) so the economics story is complete: *this page is the denominator of the data-company business case*.

12 API Sandbox — /data/api PHASE 2 & Standing Risks

Sandbox (unchanged from v1, one addition): keys with scopes, usage metering (the future billing meter), playground rendering GET /v0/maintenance?config_key=... with the Layer A+D gate applied visibly — excluded fields listed with their blocking layer. v2 addition: the playground gains a **labor endpoint preview** exposing only empirical-sourced labor (the one labor layer that is unambiguously ours), which makes the sales story concrete the day sample sizes mature.

Standing risks, named

(1) **Layer B identifiability:** whether VD-sourced rows are cleanly tagged in evidence is unverified — the audit orchestrator's Provenance Agent answers it; until then ambiguity renders X. (2) **RepairPal/MOTOR licensing:** now the high-confidence labor source, but MOTOR is sandbox-only, not contracted — its values are internal-use; the badge system must prevent them leaking into any sellable set. (3) **The 8GB VD dump:** hosting it locally is licensed use; it is not, and does not become, sellable inventory. All three converge on the same contract-review gate.

13 Schema Additions PROPOSED — WALEED & Build Plan

CHANGE	DEFINITION / NOTE
review_queue	source_stream (consensus correction report survey), entity refs, priority, status, assignee, created/resolved · thin materialization over the four real streams (§8)
data_incidents	As v1 §6; seeded with Incidents #1–2.
api_keys / api_usage	Phase 2, as v1.
audit_log extension	Confirmed existing (Jul 11) — extend with portal/entity/action/reason/diff fields as needed; do not duplicate.
notification_outbox wiring	Confirmed existing — add SLO-breach producer (§5) with per-day dedup.
Labor source tagging	Ensure labor_times.source distinguishes repairpal_motor / vehicle_databases / empirical / multiplier — required by §9.1's ladder.

PHASE	SCOPE	EXIT TEST
P0	Overview with 5 SLO tiles + Slack push · Control Room (runs, vin_queue, three triggers) · Catalog + config workspace with layer badges · Review Queue on the materialized table · Labor Command Center (ladder + test bench read-only)	"Why is this price wrong?" is answered — source ladder, evidence, fix trigger — in under two minutes, no Convex dashboard; a red SLO tile lands in Slack unaided.
P1	Parts Pricing Engine + pathology detector · Tire Intelligence · Passports + survey streams · Sources & Cache full · Verification & Variance · Coverage · Service Catalog + 7 → 4 migration workspace	The 7 → 4 cleanup executes inside the portal with a dry-run diff; a mechanic correction travels verification → queue → approved → live D-badged value entirely in-tool.
P2	Pricing Engine gated page · Provenance & Incidents · Costs & Credits · API Sandbox (post-audit, post-legal)	Layer-B exposure report matches the audit's numbers; internal key returns clean A+D responses for the top-25 seed configs; VD renewal decision is made from the credit graph.

14 Definition of Done — 10/10 Checklist

- All five SLO tiles compute from live queries against the locked thresholds; a simulated breach produces exactly one Slack notification.
- Every field value everywhere carries a computed layer badge + confidence; the consensus panel shows the real scoring math, including the $\times 1.2$ mechanic boost, on demand.
- The three director triggers (re-enrich, fix-prices, report) work per-VIN with ceremony, cooldown, and linked outcomes — verified against a test VIN end-to-end.
- Labor source ladder renders all four sources with the effective value's selection reason; the CRV/BMW regression cases are pinned and green.
- Parts median+outlier strip reproduces the May 28 worked example exactly; all three extraction pathologies detectable on seeded fixtures.
- Passport Layer 1 is structurally uneditable by any role from this portal; Layer 2 override events render with the 2–3× consistency rule visible.
- The 7 → 4 category migration completes with dry-run diff, ceremony, and zero phantom services surfacable in app search afterward.
- Four states per page, index-only queries, environment banner verified on all five deployments; no MCP-derived counts anywhere in the UI.

Otopair · Data Portal Specification v2.0 · July 11, 2026 · Supersedes v1 + Appendix A (live-validation findings incorporated inline). Grounded in: Enrichment Pipeline V8 Handoff (Mar 23) · meeting notes Apr 2, Apr 13, Apr 18 (×2), Apr 20, Apr 23, May 28, May 29, Jun 4, Jun 7, Jun 18, Jun 19, Jun 22 · Otopair Vehicle Data Pipeline monitoring spec · live production & ahmad measurements (Jul 11, Convex MCP). Companion documents: Ops Portal v1, Shops Portal v1.